

Assignment 6

Due: 5 June

This will be the last assignment of the semester.

We presented an improved version of the Contacts database program that can save contacts in a file.

I have re-uploaded the relevant files.

The file format was:

```
For each contact, each item is one line:
    name
    surname
    if email exists:
        "email"
        email
    if phone exists:
        "phone"
        phone
    if address exists:
        "address"
        address
    to mark end of contact:
        *****
```

We used “*****” to separate each contact from the next in the file. We mentioned that using separator lines in files can sometimes cause problems. Maybe we have a friend who is a super star and he decides to change his name to “*****”.

We decide to change the file format to use another common technique. Each contact will begin with the number of fields it contains. This will be a number between 2 and 5, because every contact has a first and last name and there are at most the three optional fields of email, phone and address.

We will call this new format *v2.0*. The first line of a file in the new format will be “# v2.0”.

Here is the new format specification.

```
# v2.0
For each contact, each item is one line:
  number of fields in contact (2-5)
  name
  surname
  if email exists:
    "email"
    email
  if phone exists:
    "phone"
    phone
  if address exists:
    "address"
    address
```

Modify `Contacts.java` so that it always saves in the new *v2.0* file format. It must be able to *read* contacts files in both the old and new formats.

Write a new `UseContacts.java` program that demonstrates that your program works.

Extra Credit

If you want the opportunity to earn more extra credit, do the recent lab assignment from 14 May to create a user interface for this file version of `Contacts.java`. Name your program `InteractiveContacts.java`. Extra credit will be granted at the sole discretion of the grader.

Submission

Submit a .zip file containing your files `Contacts.java`, `UseContacts.java` and the extra credit file `InteractiveContacts.java` if you wrote it.